Ziyu Shu

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Education

Georgetown University
Aug 2025 - May 2027

Master of Science in Data Science for Public Policy
Coursework: Data Science, AI and Massive Data, Relational Database Systems, Machine Learning, NLP, Causal Inference

Peking University; Visiting Student at University of California, Berkeley
Sep 2020 - Jul 2025

Bachelor of Arts in International Politics; Concentration in Computational Political Science and Conflict Prediction
Coursework: Data Structures and Algorithms, Introduction to Computing, Text Mining, Advanced Mathematics

Technical Skills

- **LLM & Agents:** LLM agents, tool use & function calling, orchestration pipelines, Claude Code, prompt engineering, versioned prompt templates, LLM-as-judge evaluation, RLHF, transformer architectures, BERT-family models, sentence embeddings, text classification
- **ML & Modeling:** machine learning, neural networks, graph neural networks (GAT), XGBoost, feature engineering, class-imbalance handling, hyperparameter tuning
- **Evaluation & Red-teaming:** evaluation pipelines, ablation harnesses, adversarial & red-team testing, jailbreak detection, model behavior analysis, lexical feature analysis, A/B testing, FPR/TPR & operating-point analysis, McNemar significance testing, AUC/ROC, ground-truth labeling
- **Engineering & Data:** Python (pandas, NumPy, PyTorch, scikit-learn, PySpark), SQL, BigQuery SQL, ETL pipelines, REST APIs, rate limiting, schema validation, session handling, retry logic
- **Full-stack & Tools:** HTML/CSS/JavaScript, dashboards, Git, Docker, MLflow, Jupyter

Work Experience

WisBot AI
New York, NY

AI Engineer
Sep 2026 - Present

- Ran end-to-end QA on an LLM agent platform that turns a product brief into a run-ready repository and GitHub workflow, reproducing failure cases across the product surface and converting them into prioritized improvement feedback.
- Improved cross-platform price-matching accuracy for the Home Depot integration: ran bulk product URLs through the Amazon matcher, classified every mismatch by root cause, then redesigned the search and matching mechanism, lifting match rate from an unstable under 20% to ~85%; shipped via feature branch and PR into development.

MLCommons
Washington, DC based Remote

AI Engineer
May 2026 - Present

- Identified past tense plus third-person reference as the grammatical signal separating narrative from non-narrative model responses, isolating it through per-sentence lexical feature analysis and weighted log-odds over 859 items; the resulting two-feature rule reached 100% precision with zero false positives and now runs in the team's production evaluation pipeline.
- Root-caused excessive false positives to LLM-scorer severity inflation after finding 95% carried no ground-truth top-severity label; built a single-command ablation harness (versioned prompt templates, stratified train/test split, automated FPR/TPR scoring with McNemar significance testing) that A/B-tested each rubric change as one controlled variable, cutting candidate FPR to 9.7%.

Massive Data Institute
Washington, DC

MDI Scholar
Jun 2026 - Present

- Built a formula-driven Excel rule-tuning tool to operationalize RuleFit-style SNAP payment-error inclusion rules across 5 household-size segments, enabling threshold adjustment without macros under restricted IT environments.

Better Government Lab
Washington, DC

Data Science Research Assistant
Sep 2025 - Present

- Reverse-engineered an undocumented public data source and built an automated collection pipeline with rate limiting and schema validation, producing 50,000+ standardized records for agency performance evaluation.

Projects

Automated LLM Job-Sourcing Agent | Python, Claude Code

- Built an autonomous, scheduled Python agent (developed with Claude Code) that scrapes LinkedIn for roles posted in the last 24 hours against multiple user-defined criteria, scores each posting for fit, and deduplicates against an applied-jobs spreadsheet to surface only new matches.
- Orchestrated the full scrape, filter, score, dedup, and schedule loop end to end, plus a daily self-maintenance pass that re-scans the results table and prunes expired or filled postings to keep output accurate without manual upkeep.

Interstate Conflict Prediction via Temporal Heterogeneous Graph Neural Networks | GitHub

- Built a PyTorch Geometric pipeline over ~200 yearly heterogeneous graph snapshots (1816–2012) from Correlates of War and Polity5, then trained a 2-layer heterogeneous GNN (HeteroConv + GAT attention) for conflict link prediction under extreme class imbalance (<2% positives), using weighted BCE loss, negative sampling, and leakage-safe chronological splits; achieved AUC 0.794 ± 0.04 across 5 seeds, outperforming a logistic-regression baseline (0.772).